

Associate Editor: Dr. Emma Sherratt

Associate Editor

Comments to the Author:

Dear authors,

I have received two expert reviews for your manuscript that indicate the scientific merits of your research, and provide directions on how to improve the manuscript.

Reviewer 1 has provided some important comments on your analytical approach that must be addressed fully in a revised manuscript. Specifically, they have identified methodological limitations to your approach of mean-centering the body size measurements, explaining with visual aids that this approach obfuscates the true patterns among the scaling relationships. You provided reasons for adopting this approach in the manuscript, but the reviewer is sceptical that this allows clear interpretation of the patterns. The reviewer has kindly provided several solutions that may be suitable to adopt instead, and I ask that you explore these with reanalysis of your data.

I request that you carefully consider all suggestions and comments made by both reviewers, and revise your manuscript accordingly.

We have addressed all reviewers' and editors' comments in detail and provided a point-by-point response (as bulleted red text) outlining how each concern was handled. We hope that the revisions meet the expectations of the reviewers and the journal.

My main point to add to theirs is that there is no imagery of the organism provided in this manuscript or supplementary materials, leaving the readers to guess the forms and structures, and so the data presented is detached from the biology.

- We've added a figure showing an individual that is representative of females and brachylabid and macrolabid males.

Comments regarding the white space of figures from Reviewer 2: for Figure 1, I appreciate you should present both females and males on the same scale y-axis. You can either cut off the female figure above 5.0, or use this space to provide visuals of

the three diet treatments, to help remind the reader of those categories.

- We believe it is important that the y-axis be on the same scale for both females and males. Therefore, we have occupied the white space with the figure legend.

Similarly, for Figure 2, visuals of the structures under study are helpful for readers unfamiliar with these organisms, and white space can be filled with such images.

Therefore, I recommend addressing Reviewer 2's concerns with some of these suggestions, especially providing visuals of the macrolabic and brachylabic forms.

A few minor comments from me:

Abstract – please provide an indication of what "forceps" are and that they are the focal "weapon".

Simply: "We quantified sex- and morph-specific weapon allometries (focusing on the forceps, which are hardened rear appendages used for defence, hunting, and mating) in the European earwig (*Forficula auricularia*) under experimental manipulation of developmental diet and density, while estimating among-family variation in intercepts and scaling exponents."

- Done

Methods – please restate that the species of study is *Forficula auricularia* in the first sentence.

- Done

TetraMin Tropical Flakes – please provide a summary of what the nutritional value of these actually is, and be explicit on how alpha cellulose affects the nutritional content.

- Done

Line 136 – suggest saying the software is the "R Statistical Environment" or "R Statistical Computing Environment", which provides more clarity.

- Done

All R packages should be cited with version used. And please provide the actual functions used as well as packages.

- Done

I look forward to receiving the revised manuscript.

Reviewer: 1

Review of Hierarchical structuring of weapon allometry in the European earwig

In this paper, the authors explore the scaling relationship between forcep length and body size (pronotum width) in female and male earwigs, under different environmental (nutrition and population density) conditions. The males come in macrolabid and brachylabid morphs, with large and small forceps, respectively, and these two forms reflect differences in mating strategy (macrolabid: fighting males; brachylabid: sneaker males). The goal of the research is to determine whether the patterns of scaling across sexes, morphs, and environmental conditions reflect changes in the intercept of the mandibles' scaling relationship with body size or in the slope of that relationship. The authors argue that this gives insight into the developmental mechanisms that regulate scaling and scaling divergence.

Several authors have explored the pattern of scaling relationships between exaggerated traits and body size, where the trait shows different scaling relationships between sexes and within sexes between males (horn scaling in *Onthophagus* beetle being the canonical example). What is particularly interesting here, however, is that the two male morphs appear to overlap considerably in overall body size, which is not clearly observed in horned beetles. This indicates that morph identity (and trait exaggeration) is not simply a reflection of overall body size. There are other systems that show this, for example, bulb mites, but it appears to be much less common.

While I think that the data are of value to researchers interested in the evolution of morphology, particularly the morphology of secondary sexual characteristics in males, I have some concerns about the paper.

Comments

1) Much of the results are concerned with differences in the intercept versus slope of the forcep-body scaling relationships among groups (males v. females, high v. low diet, macrolabid v. brachylabid, etc). The authors correctly observed that intercept and slopes in scaling relationships are typically correlated (unless a scaling relationship is rotating around the intercept, any change in slope must also change the intercept). The authors thus center the scaling relationships on mean body size (log pronotum width) per group to "facilitate interpretation of intercepts and reduce collinearity among predictors". This, however, introduces additional interpretation issues that the authors do not address.

As an example, consider the effect of diet on forcep size in macrolabid males [L258]. The authors state that "diet reduced intercept in macrolabid males [and that] Nutritional stress, therefore, disproportionately reduced weapon size in the exaggerated morph". This interpretation may be an artifact of the method of analysis. If the null hypothesis is that there is a single scaling relationship between body size and forcep size in macrolabid males, and that 'poor' diet males are smaller than 'good' diet males, then centering the scaling relationship on mean body size within each group will necessarily result in a lower intercept in the poor diet males. This can be shown in a simple illustration:

Here, red is poor diet, cyan is good diet. The left panel is the raw scaling relationship. The right panel is the scaling relationship when body size in each group is mean-centered. If the slope of the scaling relationship = 1, then there is no change in body proportion, despite an apparent change in intercept for the mean-centered scaling relationships. Thus a change in intercept the mean-centered data does not necessarily mean a change in proportion. It is true that if the slope of the scaling relationship >1, then poor diet males do have disproportionately smaller forceps; however this is captured by the slope of the scaling relationship in the uncorrected data. In short, by (unnecessarily) mean-centering the body size measurements, the authors are obfuscating the true patterns among the scaling relationships. The authors will be able to interpret their data much more easily if they do not mean center the data.

The only analysis that they will not be able to conduct is looking at the correlation between slope and intercept among

families, to test whether families with sons that had larger forceps also had steeper scaling relationships. This is because we a priori expect slopes and intercepts to be correlated. However, this could be addressed by simply correlating the slope of the family scaling relationship against mean family forcep size (figure below, left). Note, that there would also be a positive correlation if the scaling relationship between forcep and body size were not linear, but curved up (figure below, right), but this can also be tested for explicitly.

- We thank the reviewer for highlighting this important issue. We agree that group mean-centering can complicate interpretation of intercept differences and may introduce artefactual shifts when groups differ in mean body size. We have therefore reanalysed all models using uncentered log body size. This allows direct interpretation of scaling relationships and avoids conflating intercept differences with differences in mean body size.

2) A related concern is that, by centring groups on body size, the interpretation of the effects of the environment on the slopes of the scaling relationship is challenging. For example, the [L321] "size-trait relationship becomes less steep the [L321] "size-trait under a a poor diet" diet" in in males may be a consequence of non-linear scaling of the relationship between body size and forcep size, with low diet reducing body size, and hence reducing scaling. (See right panel above). Scaling relationships are a consequence of covariation in trait size as both vary in response to genetic and environmental regulators of size. The authors modify environmental factors that affect size (good diet and high density both increase body size, according to the effect in the supplemental table), but do not build this into their explanation of how these factors affect scaling. Both good diet and high density increase body size and correspondingly increase the slope of the scaling relationships: could this not be because the scaling relationship is non-linear?

- We thank the reviewer for this important suggestion. To assess whether nonlinearity could explain the observed patterns, we fitted an extended model including a quadratic term for body size. Although this model yielded a slightly higher expected log predictive density, the improvement was small relative to its standard error ($\Delta\text{elpd} = 3.7 \pm 3.8$), indicating no meaningful increase in predictive performance. In addition, credible intervals for the quadratic term overlapped zero. We

therefore found no evidence that the size–trait relationship deviates from linearity, and retained the linear model for all analyses. These results suggest that our inferences regarding differences in scaling slopes are not driven by nonlinearity.

3) Related to this, the authors report that diet and density effects on slope in males [L321], yet the credibility intervals for all these parameters include zero [L261 onward]. I would argue that there is no reasonable statistical support for this effect.

- We thank the reviewer for this important point. We agree that our original wording overstated the strength of evidence for environmental effects on scaling slopes. In the revised manuscript, we have substantially tempered our interpretation to better reflect the uncertainty in these estimates. Specifically, we now distinguish more clearly between (i) the negative body size $\times$ sex $\times$ diet interaction, whose credible interval excludes zero, and (ii) condition-specific slopes and contrasts, several of which have credible intervals overlapping zero. We therefore no longer present these results as strong evidence for slope plasticity. Instead, we describe them as providing limited or suggestive evidence that nutritional environment may influence male scaling relationships, and we explicitly note that these effects should be interpreted with caution. We have also clarified that density effects on slopes are weak and inconsistent, with credible intervals that broadly overlap zero, and we no longer interpret them as meaningful effects. These revisions are reflected in the Results (Environmental effects on scaling) and Discussion, where our conclusions now align more closely with the strength of the statistical support.

4) In numerous places the authors over-interpret their data and/or are imprecise with respect to the developmental implications of their findings. This is particularly true concerning interpretation/discussion of intercept versus slope. For example the authors argue that [L30] “difference in scaling exponent reflect variation in how strongly a trait responds to body size, whereas intercept divergence reflects differences in baseline trait expression at a given body size”. First, implicit in this statement is that trait size is a consequence of body size. However, body size is a composite measure comprising multiple traits. The developmental mechanisms that regulate covariation in trait sizes are complex, but almost certainly not as simple

as “body size determines trait size”. Second, it is not at all clear what ‘baseline’ trait expression means developmentally? Baseline relative to what? More generally, authors allude to developmental mechanisms in interpreting their data, in an imprecise manner e.g. Sex-limited expression allows selection to modify the male size–trait reaction norm without constraining female growth pathways, whereas morph differentiation primarily alters allocation levels while maintaining a shared scaling framework [L365]. What does ‘allocation’ mean developmentally, what is a ‘scaling framework’? Similarly, “differences in scaling architecture can reveal how developmental pathways partition exaggeration within a single species” [L43]. What does partition mean? What is scaling architecture? There is an increasingly good understanding of the developmental mechanisms that regulate scaling, both in model organisms such as *D. melanogaster*, and non-model horned beetles. I encourage the authors to either limit their interpretation, or embed it in a more detailed review of the developmental literature.

- We thank the reviewer for this insightful comment regarding the interpretation of scaling parameters and the need for greater precision in linking statistical patterns to developmental processes. We agree that our previous wording could be interpreted as implying overly direct or simplified developmental relationships, particularly in how we described body size and trait size, and in our use of terms such as “baseline allocation” and “scaling architecture.” In response, we have substantially revised the manuscript to clarify that our analyses describe statistical patterns of covariation rather than direct causal or mechanistic relationships.

Specifically, we have:

- Replaced language implying that trait size is determined by body size (e.g., “responds to body size”) with more neutral descriptions (e.g., “covaries with body size”).
- Removed or rephrased ambiguous or potentially misleading terms such as “baseline allocation,” “partitioning,” and “scaling architecture,” replacing them with explicit references to slope and intercept as statistical descriptors.
- Clarified the interpretation of intercepts, emphasising that they represent expected trait size at a given body size and are not directly interpretable in developmental terms in our models.
- Tempered discussion of developmental mechanisms throughout, avoiding unsupported inference and explicitly acknowledging

that our analyses do not identify the underlying developmental processes.

- Added a statement in the Discussion noting that our results quantify patterns of covariation and do not directly resolve developmental mechanisms.

These revisions substantially reduce over-interpretation and align our conclusions more closely with what can be inferred from the data. We have retained references to the developmental literature where appropriate but now frame these as possible interpretations rather than direct explanations.

We hope these changes adequately address the reviewer's concerns.

5) It is not necessary to have two analyses. A single analysis of females, brachylabial and macropalpal males is sufficient. Indeed, pooling all males together in a comparison with females, when you know that the scaling relationship between forcep and body size is different between the male morphs, renders any inferences using the pooled data unreliable.

- We thank the reviewer for this thoughtful comment regarding the structure of our analyses. We agree that male morphs differ in their scaling relationships and that treating males as a homogeneous group without accounting for this heterogeneity could lead to misleading inference.

Our intention in presenting two analyses was to address two related but distinct questions. The sex-specific model tests for overall differences between females and males in scaling relationships, including their environmental sensitivity. Importantly, this model includes interactions between body size and all predictors, allowing slopes to vary across groups and conditions. Thus, it does not assume a single common scaling relationship for all males, but rather estimates average sex differences while accounting for within-sex variation.

The morph-specific model, in contrast, focuses explicitly on variation within males by testing whether brachylabial and macropalpal individuals differ in scaling parameters and their environmental sensitivity. This allows a more detailed examination of male heterogeneity that is not the primary focus of the sex-level comparison.

To clarify this distinction and address the reviewer's concern, we have revised the manuscript to (i) more clearly justify the rationale for presenting both analyses, and (ii) explicitly acknowledge that male morph heterogeneity may influence sex-level comparisons. We now emphasise that the sex-specific results should be interpreted as capturing average differences between females and males, rather than implying that all males share a single scaling relationship.

We considered combining all groups into a single unified model including sex and morph simultaneously. However, because morph is defined only within males, this approach introduces additional complexity in model structure and interpretation (e.g., nested or conditional effects). We therefore retained the two-model approach but have clarified its purpose and limitations.

We hope this clarification resolves the reviewer's concern.

Minor points

6) The authors should present the formulae of the statistical models that they test in the MS. The results are challenging to interpret without them. The data are online, but currently hidden. I do not know whether the analysis scripts are similarly online but hidden, but should be made available.

- The statistical formulae have been added to the Supplementary Material.

7) [L317]. "Although most empirical work has focused on mean trait expression, theory does not restrict condition dependence to shifts in intercepts". I think a general interpretation of the condition dependence theory is that it pertains to the slope of scaling relationships, not just the intercepts (e.g. Bonduriansky, Am Nat, 2007, Veà et al, Am Nat, 2023).

- We've edited this section to now read: "Although many studies have focused on mean trait expression, condition dependence has also been interpreted as variation in allometric slopes (e.g. Bonduriansky 2007b; Veà et al. 2023). However, empirical tests of how environmental conditions shape scaling exponents remain relatively rare."

Reviewer: 2

Comments to the Author

Manuscript Comments to EV0-26-0122 – Hierarchical structuring of weapon allometry in the European earwig

General Comments: This was an extremely well written manuscript that I enjoyed reading. The introduction (although scattered at some points...see below) was very clear and covered all necessary information. I have minimal comments on this manuscript, with one major comment below about the figures, and the remaining comments are minor.

I find that many of the figures are poorly designed, primarily because of the massive gaps in white space in some and lack of clearly labeled axes. For example. Some figures are actually 6 panel figures which should each have their own x and y-axis. All of the key information (axes, legend, etc) are always extremely small and difficult to read.

- We thank the reviewer for bringing this to our attention. We agree and have substantially revised the figures to improve readability and clarity. In particular, we ensured that each panel has clearly labelled axes and that unnecessary whitespace is reduced. We also simplified the figure set by removing Figures 4 and 5, which were information-dense and potentially difficult to interpret, and incorporated their key results into the text. High-res versions of all figures are submitted to the journal. Their size will be adjusted subsequently, and font sizes can be modified accordingly during the proofing stage. Currently, we use 14- and 16-point font sizes for axis labels and titles, respectively.

Specific comments:

L37-38: I think before this discussion of slope-based or intercept-based you need to describe exactly what that means in the context of allometry.

- We added text to improve flow and a clearer logical progression from definition to biological meaning to examples. We reorganized the Introduction to first define slope and intercept in the context of allometry, then link these components to biological interpretation, and finally introduce the study system.

L50: Are the images of macrolabial and brachylabial forms available that can be included in this paper?

- Good idea. We've added a figure of photos of representative individuals for each sex/morph.

L48-57: I think it is strange that you introduce earwigs here (your study system), but then the following two paragraphs are more background information that does not build upon the earwig-related information you just presented. I think this can be reworked.

- We agree and have revised the Introduction to improve flow and reduce redundancy.

L86: By families, I assume you mean related individuals and not the taxonomic rank of family? I had some confusion about this.

- We have revised this section by replacing "families" with "maternal families of related individuals" at first mention.

L136: R version needs updated here.

- The version has been added.

L142: GOOD in caps.

- This has been corrected.

L144: Is a grouping factor the same as a random effect?

- Yes. We clarified that maternal identity is included as a random effect (grouping factor) in the model.

L285: Is a summary of results standard for this journal? If not, I suggest removing it.

- We have removed this section.